

Class Notes

CS & IT ENGINEERING



Theory of Computation

Turing Machine

DPP_01 Discussion Notes

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[MCQ]



#Q. Consider the following two regular expressions R_1 and R_2 over $\Sigma = \{a, b\}$.

✓ $R_1 = a^* (ba^*)^* \Rightarrow (a+b)^*$

✓ $R_2 = (a^* + b^*)^* = (a+b)^*$

Which of the following is true?

$$\frac{a^* (ba^*)^*}{(ba^*)^2} = \{ \epsilon, a^*, b^*, a^*b^*, b^*a^*, \dots \}$$

$$\begin{array}{l} ba^*ba^* \\ \hline bba^* \end{array}$$

A $R_1 \subset R_2$

B $R_2 \subset R_1$

✓ **C** $R_1 = R_2$

D $R_1 \cap R_2 = a^*b^*$

#Q. Consider the following statements:

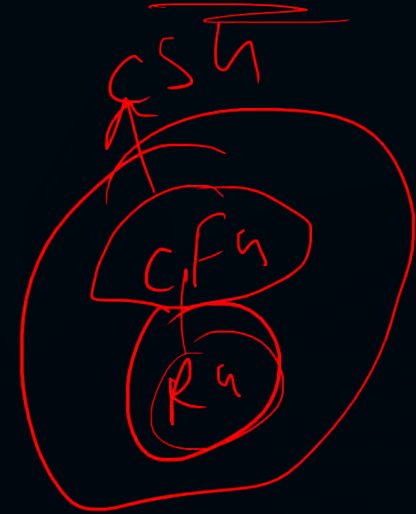
- ✓ I. A regular grammar can generate some CFL language. ✓
- ✓ II. There are some non-regular language for which PDA not exist. ✓
- ✓ III. Context sensitive grammar exists for every CFL. ✓
- ✓ IV. If L is a regular and M is not a regular language then L.M is necessarily not regular. ✓

Number of correct statement is/are _____?

3

$\{a^n b^n / n \geq 0\}$ NR $\rightarrow$ PDA
 $\{a^n b^m c^n / n \geq 0\}$ CSL $\rightarrow$ Non-Reg $\rightarrow$ LBA

$\phi \cdot \{a^n b^n / n \geq 0\}$
 $\Rightarrow \phi$



#Q. Which of the following languages are not decidable?

A $L = \{ \langle G \rangle \mid G \text{ is CFG and } R \text{ is a regular set such that } L(G) = R \}$

UD

CFL

B $L = \{ \langle G \rangle \mid G \text{ is CFG such that } L(G) = \phi \}$

Decidable

mem ✓

C $L = \{ \langle G_1, G_2 \rangle \mid G_i \text{ is CFG such that } L(G_1) \cap L(G_2) = \Sigma^* \}$

completeness ✓
Finiteness ✓
EMP ✓

D $L = \{ \langle G_1, G_2 \rangle \mid G_i \text{ is CFG such that } L(G_1) = L(G_2) \}$

UD

UD

#Q. Consider the following problems:

P_1 : Does a given program ever produce an output? $\rightarrow$ *Non-Trivial*

P_2 : Given DFAs D_1 and D_2 is $L(D_1) \cap L(D_2) = \emptyset$? *Decidable*

P_2 : For Context sensitive grammar G and string w , is $w \in L(G)$? *membership*

How many languages is/are Undecidable? ____.

1

decidable

[MCQ]



#Q. If $L_1 \cap L_2$ is regular and L_1 is non-regular then, L_2 must be

$L_1 \cap L_2 = \text{Reg}$
 $\text{NR} \cap L_2 = \text{Reg}$

$\left. \begin{array}{l} \{a^n b^n\} \cap \phi = \phi \\ \{a^n b^n\} \cap \{a^n b^n\} = \{a^n b^n\} = \text{Reg} \end{array} \right\}$

ϕ

A Regular but not finite $\times$

B Non-regular $\times$

C Finite $\times$

D None of these

$\{a^n b^n \mid n \geq 0\}$ $\cap$ $\{a^n b^n \mid n \geq 0\}$

L_1 ϕ

$P_1 \leq P_2$

#Q. If P_1 is reducible to P_2 then, which of the following is/are correct?

A

If P_1 is decidable then, P_2 is undecidable. *we can't say* X

B

If P_2 is undecidable then, P_1 is decidable. X

C

If P_2 is decidable then, P_1 must be decidable. ✓

D

If P_1 is undecidable then, P_2 must be undecidable. ✓

#Q. Let,

✓ $L_1 = \text{CFL}$

✓ $L_2 = \text{DCFL}$

✓ $L_3 = \text{Regular}$

Then, which of the following is/are correct?

B, D

$$\underline{L_1 - L_2} = L_1 \cap \overline{L_2}$$

$$a^n b^n \cdot \phi = R$$

$$\underline{a^n b^n \cdot c} = \text{DCFL}$$

- A** ✓ $L_3 - L_1$ is CSL $\Rightarrow \text{Reg} \cap \overline{\text{CFL}} \Rightarrow \text{Reg} \cap \text{CSL} \Rightarrow \underline{\text{CSL}}$
- B** ✓ $L_1 \cup (L_2 \cap \overline{L_3})$ is CFL $\Rightarrow \text{CFL} \cup (\text{DCFL} \cap \overline{\text{Reg}}) = \text{CFL} \cup \text{DCFL} = \text{CFL}$
- C** $L_2 \cdot L_3$ is Regular $= \underline{\text{DCFL}} \cdot \text{Reg} = \text{X}$
- D** ✓ $L_2 \cup L_3$ is DCFL $\text{DCFL} \cup \text{Reg} = \text{DCFL}$

#Q. Consider the following statements:

S1: Complement of finite language may be finite. ~~X~~

S2: Kleene star of finite language may be finite. ✓

S3: Subset of finite language is always finite. ✓

S4: Complement of infinite language may be finite. ✓

Number of INCORRECT statements is/are ____.

1

$$\begin{aligned}\overline{\Phi} &= (a+b)^* \\ \epsilon &= \overline{(a+b)^* - \epsilon} \\ S2 \Rightarrow \Phi^* &= \epsilon \\ \epsilon^* &= \epsilon\end{aligned}$$

$$\{ \underline{ab, ba, aa, bbb} \}$$

✓✓

#Q. Which of the following problems are undecidable?

B, D

A

~~Membership problem in context-free languages.~~

B

~~Whether a given context-free language is regular.~~

C

~~Whether a finite state automation halts on all inputs.~~

D

~~Membership problem for type-0 languages.~~

CFL

$\Rightarrow M$

$\Rightarrow F$

$\Rightarrow E$

RE

R_n

CSL

CFL

R_0

FSM

Type-0

Type-1

Type-2

Type-3

RE

[MCQ]



#Q. Which one of the following is UNDECIDABLE?

B

- ☒ **A** *Decidable* Given a Turing machine M, a string s and an integer k, M accepts s within k steps
- ☒ **B** Equivalence of two given Turing machines *UD*
- ☒ **C** Language accepted by a given finite state machine is not empty *Decidable*
- ☒ **D** Language generated by a context free grammar is non empty *Decidable*

[MCQ]



#Q.

Consider the following decision problems:

(P₁) Does a given finite state machine accept a given string? ✓

(P₂) Does a given context free grammar generate an infinite number of strings

Which of the following statements is true?

A

Both (P₁) and (P₂) are decidable ✓

B

Neither (P₁) nor (P₂) are decidable

C

Only (P₁) is decidable

D

Only (P₂) is decidable

[MCQ]



#Q. Which of the following problems is undecidable?

A Membership problem for CFGs. Dec

B Ambiguity problem for CFGs. UD

C Finiteness problem for FSAs. Dec

D Equivalence problem for FSAs. Dec

<u>CFL</u>	<u>Reg</u>	<u>CSL</u>
F	✓	X
m	✓	✓
E	✓	X

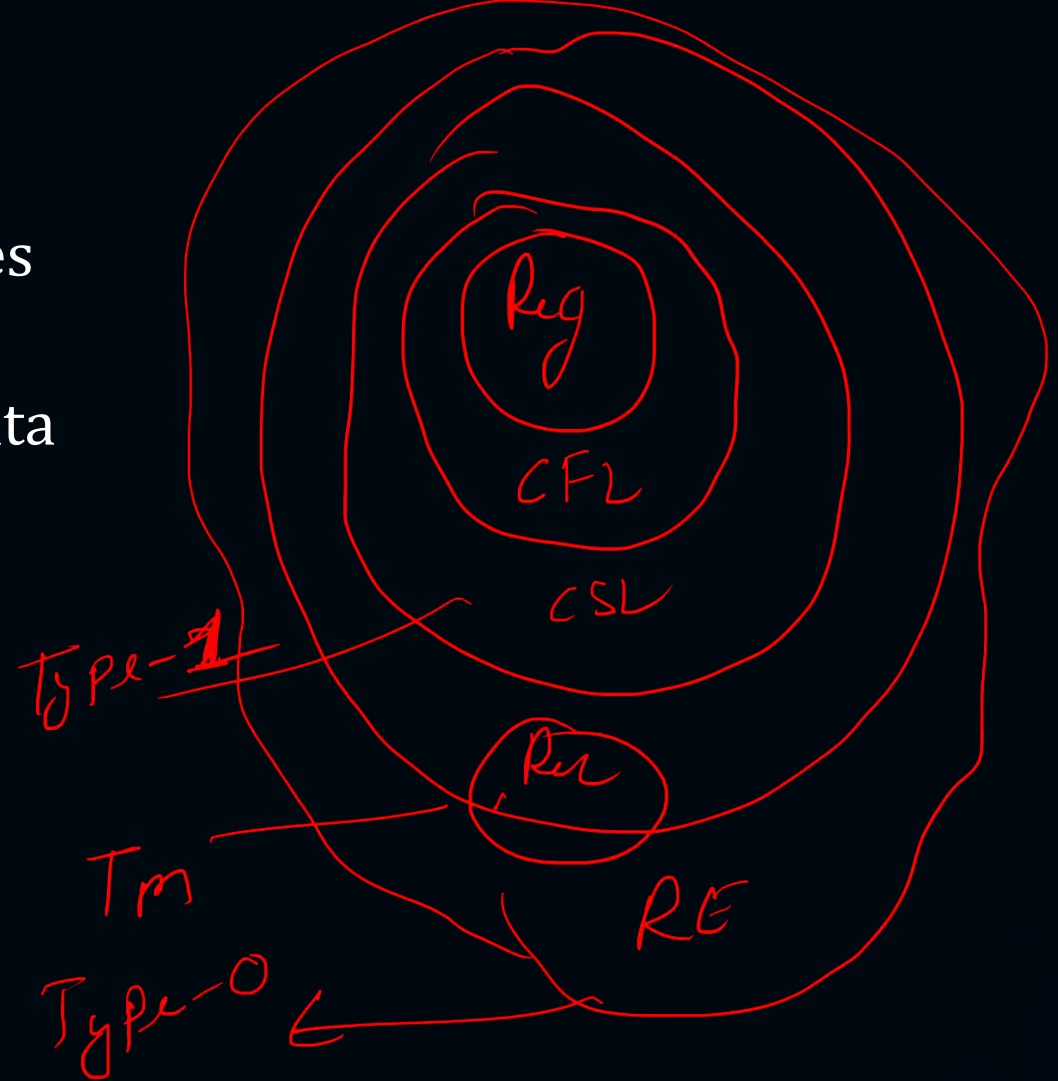
#Q. Recursive languages are:

A a proper superset of context free languages

B ~~always~~ recognized by a pushdown automata

C also called type-0 languages

D recognizable by Turing machines



Classics over

Telegram

SWT $\Rightarrow$ 50-60 problems
 3DPP $\Rightarrow$ 40-50 problems
 CN $\Rightarrow$ 200 problems
 $\Rightarrow$ 100 problem

THANK - YOU

300
250 PYQ

4-6 $\rightarrow$ Reg $\rightarrow$ 50% 60% { Closure properties }

2-4 $\rightarrow$ CFG $\rightarrow$ 25-30%

TM $\Rightarrow$ 1 2 marks

DFA $\rightarrow$ RC, RS

SSO